INTRODUCTION

In this report, we will compare the economic performance of Switzerland and Latvia between 2012 to 2023. We'll take a close look at key indicators like GDP, price levels, unemployment rates, and trade openness to uncover the differences in growth patterns, economic structures, and productivity between these two nations. We'll also examine how factors such as human capital, capital accumulation, and the quality of institutions have influenced economic outcomes.

TASK 1

We can observe how each one of the countries analysed which are Switzerland and Latvia have different values of Real GDP and Nominal GDP. This fundamentally happens because Nominal GDP measures the economic growth considering the change in the quantity of output and also the change in prices, while the Real GDP measures the economic growth taking into account only the change in output, keeping prices constant (in this case at 2015 prices). Subsequently we can also analyse which one of the two countries has grown more in nominal and real terms. Firstly, we can observe how, in nominal terms, Latvia's GDP has grown from 21.270,90 million euros to 39.564,20 million euros (from 2012 to 2023) which could be translated as a growth rate of 86,00%. On the other hand, in Real terms (at 2015 prices) we can see how the nation's GDP has grown from 21.949,13 million euros to 27.553,97 million euros (from 2012 to 2023) which could be translated as a growth rate of 25,53%. As we can predict, the growth in real terms is lower than the one in nominal terms. Secondly, we can observe how, in nominal terms, Switzerland's GDP has grown from 534.012,80 to 826.951,80 (from 2012 to 2023) which could be translated as a growth rate of 54,86%. On the other hand, in real terms (at 2015 prices) we can see how the country's GDP has grown from 590.695,99 to 721.566,95 which could be translated as a growth rate of 22,16%. Thus, we can conclude that although Switzerland has the largest total GDP, Latvia has experienced a higher growth rate both in nominal and real values of GDP.

With regards to the price growth, we use two price indexes, the CPI and the GDP Deflator. The CPI (Consumer price Index) measures the evolution of the prices of a fixed basket of goods that the average household of a country would consume. The GDP Deflator measures the overall price of all that is produced in the economy. Firstly, we can observe how Latvia's CPI has grown from 99,09% to 143,38% which could be translated as a growth rate of 44,69%. While its GDP Deflator has grown from 96,91% to 143,59% which translates to a growth rate of 48,17%, relatively higher than the CPI. Secondly, we can see how Switzerland's CPI has grown from 100,76% to 106,10% which indicates a growth rate of 5,30%. On the other hand, the GDP Deflator has increased from 90,40% to 114,61% which indicates a growth rate of 26,78%. As we can observe, considering both the CPI and the GDP Deflator, prices grew more in Latvia compared to Switzerland.

Moving onto the next aspect, it can be noticed how in 2020, for Latvia, the unemployment increased to 8,10% while the country had a decrease of -3,47% of real GDP. A similar scenario unfolded with Switzerland, where unemployment increased to 4,80% while GDP in real terms decreased by -2,14%. On the other hand, we can also observe how, after the COVID crisis, in 2021 Latvia's GDP and Switzerland's GDP increased by 6,94% and 5,56%

respectively. However, while Latvia's unemployment did decrease to 7,60%, Switzerland's unemployment increased to 5,10%. We can conclude that indeed there is an empirical relationship between unemployment and GDP that suggests that the higher the unemployment the lower will be the GDP, however, in this case this relationship is stronger in Latvia. The situation of Switzerland could have been a result of several factors. Since 2021 was a transition year after the COVID crisis, people re-entered the market, which provoked the labour supply to increase faster than the labour demand. Another factor could be the fact that the Swiss economy relies majority on capital-intense companies who do not need a large number of human capital. So therefore, while GDP did grow, unemployment also grew.

With respect to this period of time, between 2020 and 2021, we can also observe a notable change in other variables like price indexes. For instance, in Latvia, inflation was very low with an increase of CPI of 0,08% during the COVID crisis in 2020 while in Switzerland we had a staggering deflation of -0,84%. On the other hand, GDP Deflator did grow, showing an increase in prices of 2,39% in Latvia and an increase of 3,19% in Switzerland. Based on the data shown, we can conclude that in 2020 consumer prices fell or barely grew, while the prices of the goods that the economy produced did not fall but rather grew. During the following year, Latvia saw the start of higher inflation, with an increase of CPI and Deflator of 3,24% and 3,30% respectively. Switzerland also experienced an inflation of 0,48% according to CPI and of 0,33% according to the GDP Deflator, which is relatively lower than Latvia.

TASK 2

To begin with, it may be observed through the data elaborated, how much each GDP component weights relatively with respect to the total aggregate GDP. For instance, In Latvia household consumption consistently ranks as the highest contributor to total GDP with an annual rate of 58%-61% over the period (2012-2023). Government spending shows stability, with rates between 18%-21%. Gross capital formation shows a little bit more of fluctuations with rates varying between 24%-32% but still keeping a pretty consistent change. However, the most notable feature of Latvia's GDP structure is the high external sector: imports represent about 64%-84% of GDP reaching peak numbers like 84% in 2022, while exports make up roughly 60%-74% of GDP. In contrast, Switzerland allocates a smaller share of household consumption (around 49%-53%), government consumption also stays pretty consistent with rates around (11%-12%) while relatively lower than Latvia. Gross capital formation appears relatively constant (around 24%-27%). Like Latvia, Switzerland also shows a big portion of GDP being compromised by exports and imports making up about 60%-69% and 51%-55% respectively.

The volatility of a component of GDP could be measured obtaining its corresponding standard deviation. Firstly, for Latvia, we can observe how the annual rate of gross capital formation is the one with the highest volatility with a standard deviation of (11), followed by both exports and imports (5), then the annual rate of household consumption expenditure (4) and finally the one with the lowest is the annual rate of general government consumption expenditure (2). Thus, we can conclude that gross capital formation has the highest standard deviation of annual growth rates which indicates that its growth fluctuates the most over time. In contrast, for Switzerland, we can see how the annual rate of exports of goods and services is the one with the highest volatility with a standard deviation of (6), followed by imports (5), then the annual rate of gross capital formation (4) then the annual rate of

household consumption expenditure (2) and finally the one with the lowest is the annual rate of general government consumption expenditure (1). As we can conclude, exports of goods and services have the highest standard deviation of annual growth rates which indicates that its growth fluctuates the most over time.

The trade openness index of a country measures how open an economy is to international trade. Initially, one can observe how Latvia's trade openness index is higher than Switzerland's index. According to the data shown, Latvia's trade openness has been fluctuating over the period, starting at 128% in 2012 with a marked rise after 2019 achieving peaks of 158% in 2022. By contrast, Switzerland shows more stable values. While it is true that it has been also fluctuating over the period, the changes are much lower compared with Latvia, with values ranging between 111% and 124% and without a clear upward pattern. So therefore we can conclude that Latvia has a more open economy than Switzerland, characterised by higher and faster-growing trade openness, while Switzerland's trade trajectory is more stable and moderate.

TASK 3

According to the 2017 data elaborated, we are going to analyse whether the Solow-Swan predictions are fulfilled. The Solow-Swan model (1956) is a model that primarily focuses on the importance of physical capital accumulation, together with population growth and technological status, to explain why some countries achieve high levels of wealth in the long run, while others do not. Firstly, for Switzerland, we can observe that its capital stock per worker is substantially higher than Latvia's (541.532 vs 409.372), and this corresponds to a much higher level of output per worker (124.512 vs 52.996). This strong positive relationship supports the model's prediction that more capital intensification raises productivity. However, the expected negative relationship between population growth and output per worker does not appear clearly in this scenario. For instance, Latvia's population growth is negative (-1,17%) while Switzerland growth is positive (+0,91%) despite this, Switzerland still depicts a higher capital stock and a higher level of GDP per worker. This deviation from the prediction may be linked to several factors such as higher investment share, greater human capital or better quality of governance that offset the effect of growing population rate. As a result, we can conclude that while the prediction between capital and output is aligned with the model, the influence of population growth is diminished by other factors in each country.

Based on the information developed, one can clearly see the relationship between GDP per worker and several variables. First, countries with higher human capital (hc) tend to achieve higher productivity: Switzerland, with an index of 4, shows a much higher output per worker than Latvia, whose index is 3. In contrast, total natural resources do not show a consistent positive relationship, for example, Latvia's resource rents which are significantly larger (1,207) coincide with a lower output per worker than Switzerland whose natural resources are relatively lower (0,009). This result indicates that having abundant natural resources alone does not automatically translate into higher income. Regarding corruption we can observe that Switzerland scores higher in the Corruption Perceptions Index (82) compared to Latvia who scores 60 over 100. Now this can be linked with the fact that Switzerland has a higher GDP per worker (124,512) in comparison to Latvia's GDP per worker (52,996). This pattern may suggest a positive relationship between income and institutional quality: countries with higher income tend to have lower levels of corruption. Finally, rule of law also

shows a strong link with productivity: Switzerland's significantly higher score (99,05) with respect to Latvia’s score (78,57-80) aligns with the principle that a higher institutional quality has a positive effect on a country's income. In conclusion, we can deduce that the higher the human capital and the stronger the institutional quality in a country, the higher will be the GDP per worker.

In summary, comparing Switzerland and Latvia reveals that their economic performances are significantly influenced by differences in human capital, the quality of institutions, and how much capital they accumulate. Switzerland boasts a higher overall GDP and productivity, but Latvia has been growing at a quicker pace, especially when looking at both nominal and real GDP. Additionally the GDP structure, the trade openness and how unemployment responds to economic shocks also play a crucial role in shaping economic impacts and long-term development.